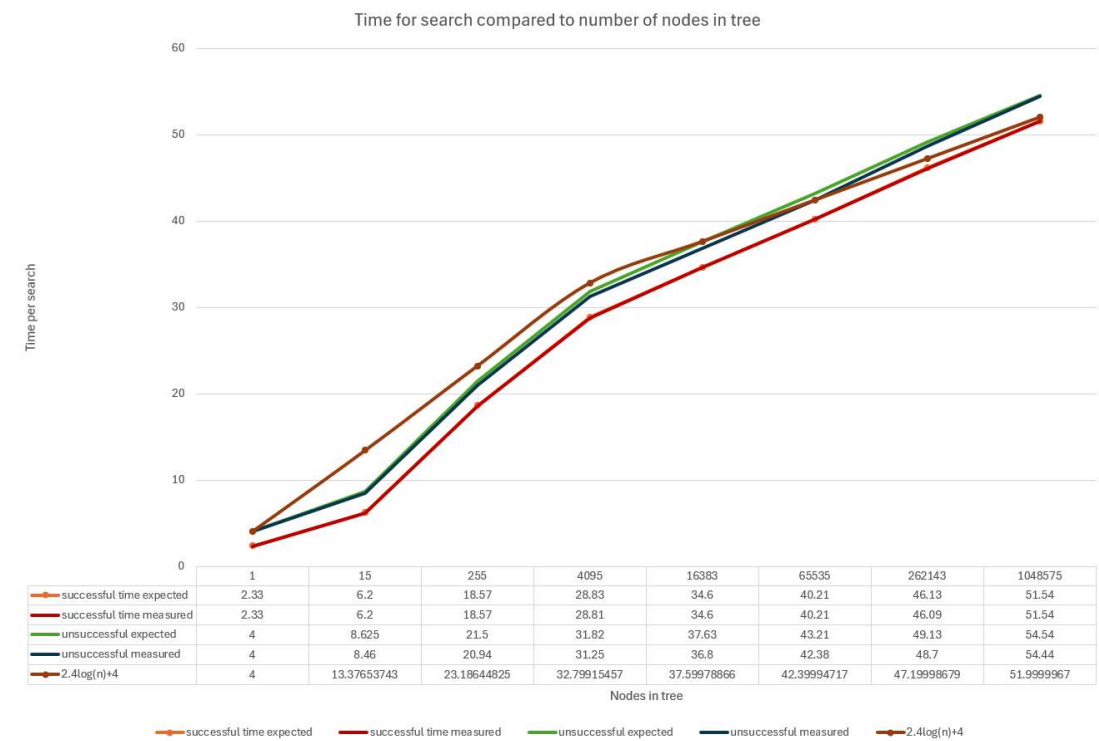
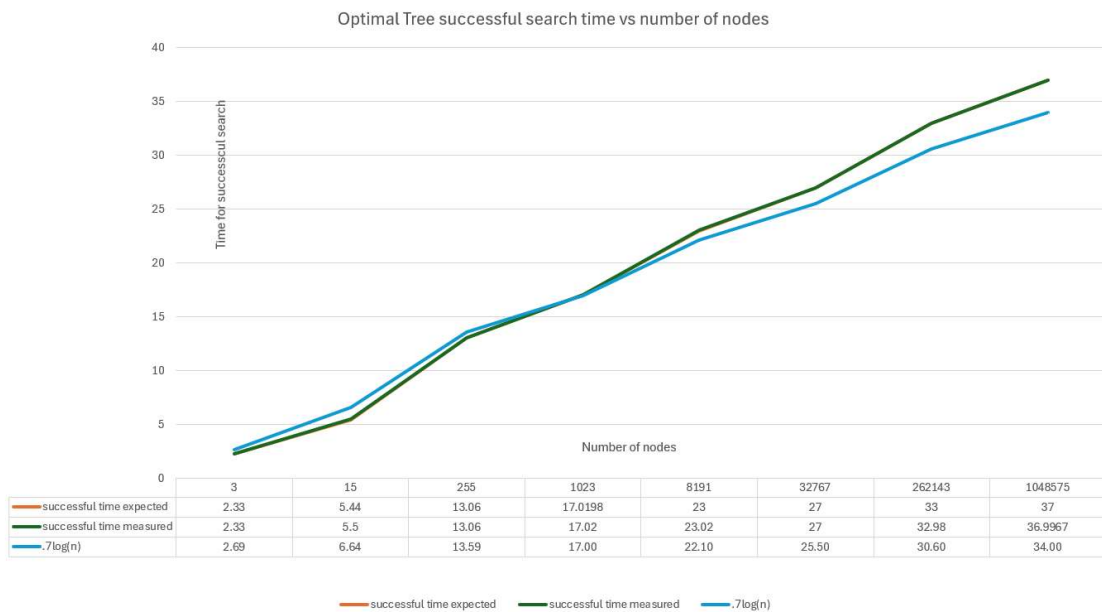


Performance analysis:
Drew Buley
11/13/2024
MP5

This graph shows the search times for successful and unsuccessful searches compared to their measured values for tree depths of max level 1 to max level 20. Note that for the successful search, my values nearly directly matched the expected so the lines are overlaid.



When comparing to the optimal tree layout, we can observe that the graph for the expected search time is the following:



By analyzing the trendline formulas for these two graphs, we can see that the line of best fit is derived from an adjusted log base 2 formula, and that by using an optimal tree our search efficiency improved by roughly 242%. Because of this, it is easy to claim that this code is of time complexity $O(\log n)$. Still, both of these trees are quite faster than a tree functioning in the worst case (I.E in form similar to a linked list) which functions in $O(n)$ time.